

ASSINGMENT--6 (RESPONSIBLE AI)

Task Description #1 (Privacy & Security in File Handling)

PROMPT: generate a Python script that stores user data (name, email, password) in a file. AND Revised version with encrypted password storage (e.g., hashing).

CODE:

```
def store_user_data(name, email, password):

    # Open the file in append mode

    with open('user_data.txt', 'a') as file:

        # Write the user data to the file

        file.write(f'Name: {name}, Email: {email}, Password: {password}\n')

#example usage

if __name__ == "__main__":

    # Get user input

    name = input("Enter your name: ")

    email = input("Enter your email: ")

    password = input("Enter your password: ")

    # Store the user data

    store_user_data(name, email, password)

    print("User data stored successfully.")

#generate a Revised version with encrypted password storage (e.g., hashing).

# Python script to store user data (name, email, password) in a file with encrypted
password storage

import hashlib
```

```
def hash_password(password):  
    return hashlib.sha256(password.encode()).hexdigest()  
  
def store_user_data(name, email, password):  
    # Hash the password  
    hashed_password = hash_password(password)  
  
    # Open the file in append mode  
    with open('user_data.txt', 'a') as file:  
        # Write the user data to the file  
        file.write(f'Name: {name}, Email: {email}, Password: {hashed_password}\n')  
  
#example usage  
if __name__ == "__main__":  
    # Get user input  
    name = input("Enter your name: ")  
    email = input("Enter your email: ")  
    password = input("Enter your password: ")  
  
    # Store the user data  
    store_user_data(name, email, password)  
  
    print("User data stored successfully.")
```

OUTPUT:

```
PS C:\Users\vishal\Documents\.vscode> python -u "c:\Users\vishal\Documents\.vscode\github\tempCodeRunnerFile.py"
Enter your name: vishal
Enter your email: vishal@gmail.com
Enter your password: vishal@16
User data stored successfully.
Enter your name: vishal
Enter your email: vishal@gmail.com
Enter your password: vishal@16
User data stored successfully.
PS C:\Users\vishal\Documents\.vscode> []
```

Task Description #2 – Fair Decision Logic

PROMPT: generate a scholarship eligibility checker based on academic score, family income, and location. Examine whether the logic unfairly favors or disadvantages any group.

CODE:

```
def check_scholarship_eligibility(academic_score, family_income, location):
```

```
    # Define eligibility criteria
```

```
    min_academic_score = 85
```

```
    max_family_income = 50000
```

```
    eligible_locations = ['urban', 'suburban']
```

```
    # Check academic score
```

```
    if academic_score < min_academic_score:
```

```
        return "Not eligible: Academic score is below the minimum requirement."
```

```
    # Check family income
```

```
    if family_income > max_family_income:
```

```
        return "Not eligible: Family income exceeds the maximum limit."
```

```
# Check location

if location.lower() not in eligible_locations:

    return "Not eligible: Location does not meet the eligibility criteria."


return "Eligible for scholarship."

# Example usage

if __name__ == "__main__":

    # Get user input

    academic_score = float(input("Enter your academic score (0-100): "))

    family_income = float(input("Enter your family income: "))

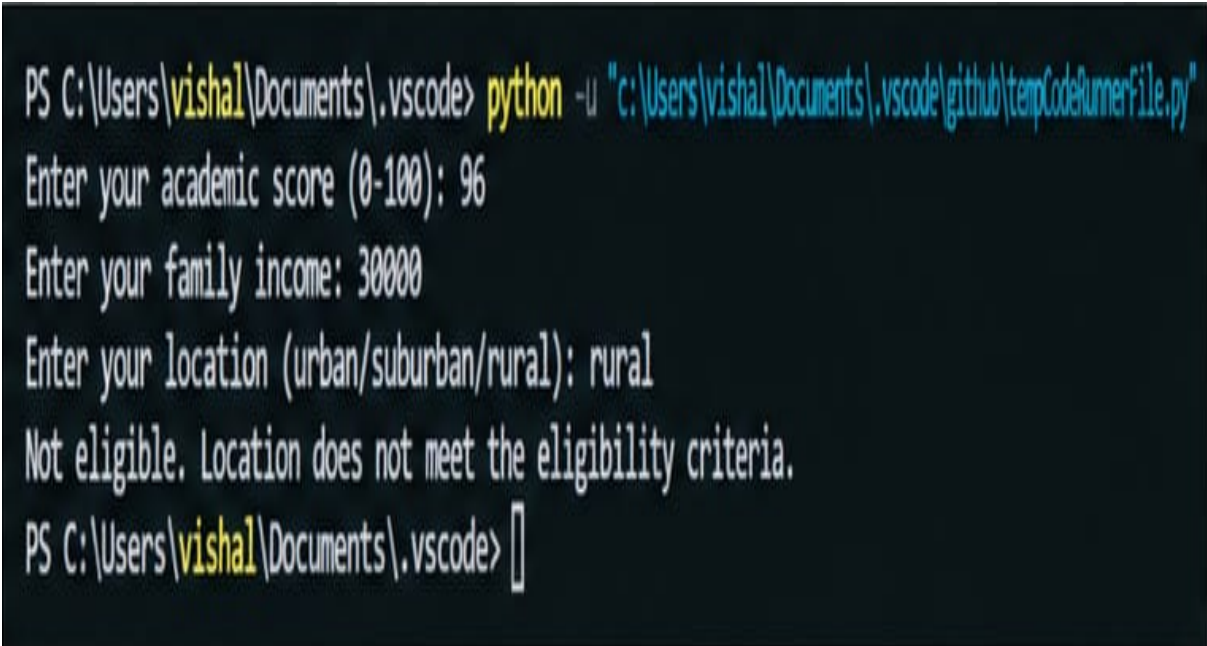
    location = input("Enter your location (urban/suburban/rural): ")


    # Check scholarship eligibility

    result = check_scholarship_eligibility(academic_score, family_income, location)

    print(result)
```

OUTPUT:



```
PS C:\Users\vishal\Documents\.vscode> python -u "c:\Users\vishal\Documents\.vscode\github\tempCodeRunnerFile.py"
Enter your academic score (0-100): 96
Enter your family income: 30000
Enter your location (urban/suburban/rural): rural
Not eligible. Location does not meet the eligibility criteria.
PS C:\Users\vishal\Documents\.vscode> 
```

Task Description #3 (Transparency in AI-Based Admission System)

PROMPT: generate a Python program that predicts whether a student should be selected or rejected for admission based on factors such as entrance exam score, academic marks, and other relevant admission criteria.

CODE:

```
def predict_admission(entrance_exam_score, academic_marks,
extracurricular_activities):

    # Define admission criteria

    min_entrance_exam_score = 70

    min_academic_marks = 75

    extracurricular_weight = 10

    # Calculate total score

    total_score = entrance_exam_score + academic_marks + extracurricular_activities *
extracurricular_weight

    # Determine admission decision

    if entrance_exam_score < min_entrance_exam_score:

        return "Rejected: Entrance exam score is below the minimum requirement."

    if academic_marks < min_academic_marks:

        return "Rejected: Academic marks are below the minimum requirement."

    if total_score >= 200:

        return "Selected for admission."

    else:
```

```

        return "Rejected: Total score does not meet the admission criteria."

# Example usage

if __name__ == "__main__":

    # Get user input

    entrance_exam_score = float(input("Enter your entrance exam score (0-100): "))

    academic_marks = float(input("Enter your academic marks (0-100): "))

    extracurricular_activities = int(input("Enter the number of extracurricular activities
participated in: "))

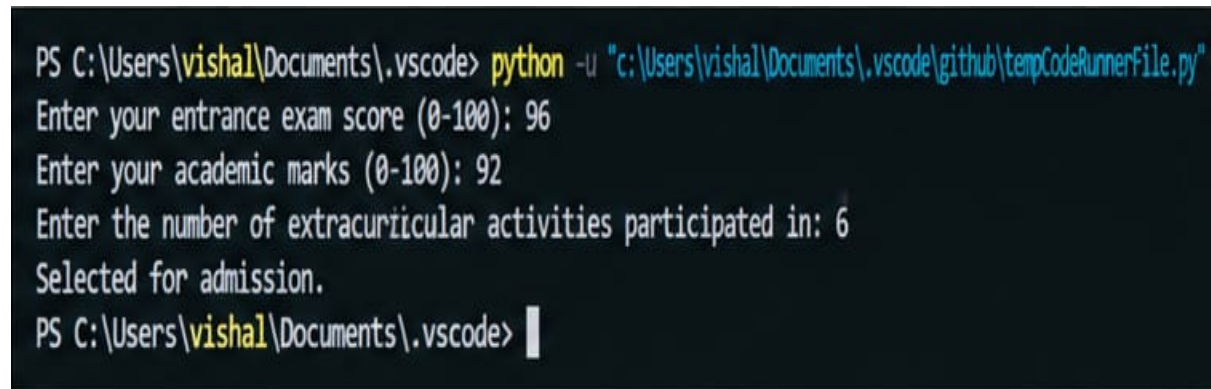
    # Predict admission decision

    result = predict_admission(entrance_exam_score, academic_marks,
extracurricular_activities)

    print(result)

```

OUTPUT:



```

PS C:\Users\ Vishal\Documents\.vscode> python -u "c:\Users\ Vishal\Documents\.vscode\github\tempCodeRunnerFile.py"
Enter your entrance exam score (0-100): 96
Enter your academic marks (0-100): 92
Enter the number of extracurricular activities participated in: 6
Selected for admission.
PS C:\Users\ Vishal\Documents\.vscode>

```

Task Description #4 – Ethical Scoring System

PROMPT: Task: Generate an employee performance evaluation system using inputs such as project completion rate, teamwork score, and attendance. Analyze the scoring logic for any unethical weighting or hidden bias.

CODE:

```

def evaluate_employee_performance(project_completion_rate, teamwork_score,
attendance):

```

```
# Define performance criteria

min_project_completion_rate = 70

min_teamwork_score = 60

min_attendance = 80


# Calculate total score

total_score = (project_completion_rate * 0.5) + (teamwork_score * 0.3) + (attendance
* 0.2)


# Determine performance evaluation

if project_completion_rate < min_project_completion_rate:

    return "Needs Improvement: Project completion rate is below the minimum
requirement."


if teamwork_score < min_teamwork_score:

    return "Needs Improvement: Teamwork score is below the minimum requirement."


if attendance < min_attendance:

    return "Needs Improvement: Attendance is below the minimum requirement."


if total_score >= 75:

    return "Excellent Performance."

elif total_score >= 60:

    return "Good Performance."

else:

    return "Needs Improvement."

#example usage

if __name__ == "__main__":

    # Get user input
```

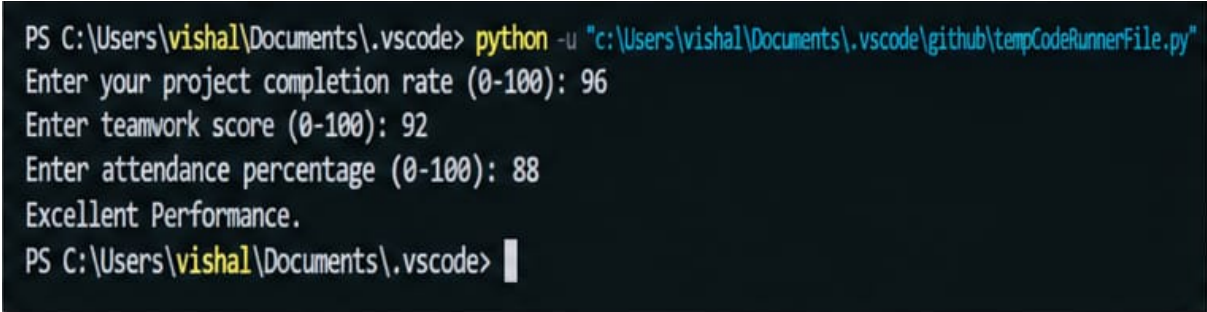
```
project_completion_rate = float(input("Enter project completion rate (0-100): "))
teamwork_score = float(input("Enter teamwork score (0-100): "))
attendance = float(input("Enter attendance percentage (0-100): "))

# Evaluate employee performance

result = evaluate_employee_performance(project_completion_rate, teamwork_score,
attendance)

print(result)
```

OUTPUT:



```
PS C:\Users\vishal\Documents\.vscode> python -u "c:\Users\vishal\Documents\.vscode\github\tempCodeRunnerFile.py"
Enter your project completion rate (0-100): 96
Enter teamwork score (0-100): 92
Enter attendance percentage (0-100): 88
Excellent Performance.
PS C:\Users\vishal\Documents\.vscode>
```

Task Description #5 – Accessibility and Inclusiveness

PROMPT: generate a user feedback form application. Review whether the language and options provided are inclusive and accessible to diverse users.

CODE:

```
def user_feedback_form():

    # Collect user feedback

    print("Welcome to the User Feedback Form")

    name = input("Please enter your name: ")

    email = input("Please enter your email: ")

    feedback = input("Please provide your feedback: ")


# Store feedback in a file

with open('user_feedback.txt', 'a') as file:
```

```
file.write(f'Name: {name}, Email: {email}, Feedback: {feedback}\n')
```

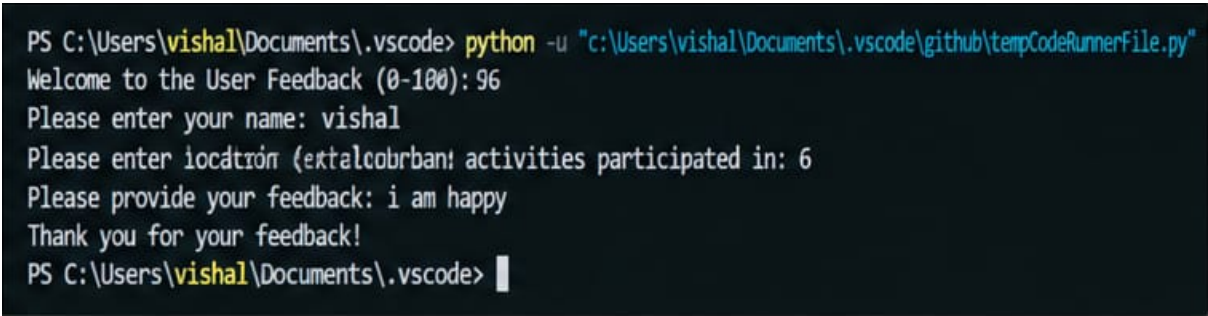
```
print("Thank you for your feedback!")
```

```
#example usage
```

```
if __name__ == "__main__":
```

```
    user_feedback_form()
```

OUTPUT:



```
PS C:\Users\vishal\Documents\.vscode> python -u "c:\Users\vishal\Documents\.vscode\github\tempCodeRunnerFile.py"
Welcome to the User Feedback (0-100): 96
Please enter your name: vishal
Please enter location (extalcourban: activities participated in: 6
Please provide your feedback: i am happy
Thank you for your feedback!
PS C:\Users\vishal\Documents\.vscode> |
```

Task Description #6 (Fairness in Loan Approval System)

PROMPT: generate a Python program that predicts whether a person should be approved or rejected for a loan based on information such as income, age, employment status, credit score, and loan amount.

CODE:

```
def predict_loan_approval(income, age, employment_status, credit_score,
loan_amount):
```

```
    # Define loan approval criteria
```

```
    min_income = 30000
```

```
    min_age = 21
```

```
    min_credit_score = 650
```

```
    max_loan_amount = 50000
```

```
    # Check income
```

```
    if income < min_income:
```

```
        return "Rejected: Income is below the minimum requirement."
```

```
# Check age

if age < min_age:

    return "Rejected: Age is below the minimum requirement."


# Check employment status

if employment_status.lower() not in ['employed', 'self-employed']:

    return "Rejected: Employment status does not meet the eligibility criteria."


# Check credit score

if credit_score < min_credit_score:

    return "Rejected: Credit score is below the minimum requirement."


# Check loan amount

if loan_amount > max_loan_amount:

    return "Rejected: Loan amount exceeds the maximum limit."


return "Approved for loan."


# Example usage

if __name__ == "__main__":

    # Get user input

    income = float(input("Enter your income: "))

    age = int(input("Enter your age: "))

    employment_status = input("Enter your employment status (employed/self-employed/unemployed): ")

    credit_score = int(input("Enter your credit score (300-850): "))

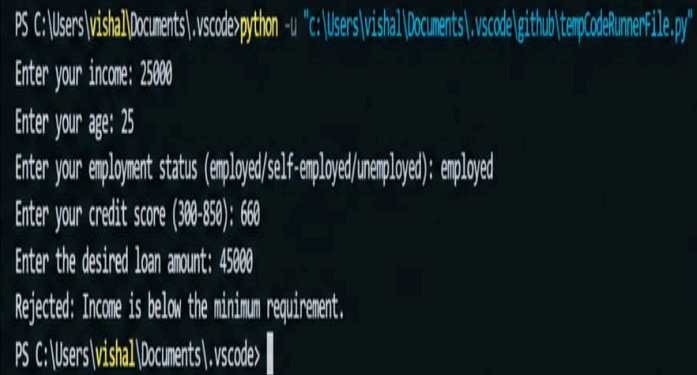
    loan_amount = float(input("Enter the desired loan amount: "))
```

```
# Predict loan approval

result = predict_loan_approval(income, age, employment_status, credit_score,
loan_amount)

print(result)
```

OUTPUT:

A screenshot of a Windows PowerShell terminal window with a dark background. The prompt is 'PS C:\Users\vishal\Documents\.vscode>'. The user enters 'python -u "c:\Users\vishal\Documents\.vscode\github\tempCodeRunnerFile.py"'. The program then prompts for user input: 'Enter your income: 25000', 'Enter your age: 25', 'Enter your employment status (employed/self-employed/unemployed): employed', 'Enter your credit score (300-850): 660', and 'Enter the desired loan amount: 45000'. The program outputs 'Rejected: Income is below the minimum requirement.' and returns to the prompt 'PS C:\Users\vishal\Documents\.vscode>'.

```
PS C:\Users\vishal\Documents\.vscode>python -u "c:\Users\vishal\Documents\.vscode\github\tempCodeRunnerFile.py"
Enter your income: 25000
Enter your age: 25
Enter your employment status (employed/self-employed/unemployed): employed
Enter your credit score (300-850): 660
Enter the desired loan amount: 45000
Rejected: Income is below the minimum requirement.
PS C:\Users\vishal\Documents\.vscode> |
```